

VADEROARNA, Sweden

WMO: 025010

Lat: 58.576N

Lon: 11.066E

Elev: 24

StdP: 101.04

Time Zone: 1.00 (EUC)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-9.6	-7.5	-13.2	1.2	-8.1	-11.3	1.4	-5.4	19.2	4.0	17.8	4.1	7.3	60	0.855

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.0	22.9	17.9	21.3	17.6	20.1	17.0	19.1	21.6	18.3	20.4	17.5	19.5	5.1	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	13.1	20.4	17.4	12.5	19.6	16.6	11.9	18.8	54.6	21.7	52.0	20.5	49.4	19.5	21.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.2	16.4	15.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.6	0.6	-0.1	2.4	6.4	11.3	14.9	17.5	16.9	14.2	9.7	5.7	2.7
	DBStd	6.82	3.82	3.74	2.78	2.62	3.22	2.28	2.14	1.59	2.01	2.46	3.50	4.39
	HDD10.0	1335	291	282	236	114	19	0	0	0	1	36	129	226
	HDD18.3	3591	549	516	494	359	219	106	43	50	124	268	378	484
	CDD10.0	813	0	0	0	5	61	148	231	214	127	26	2	0
	CDD18.3	27	0	0	0	0	1	4	16	6	0	0	0	0
	CDH23.3	29	0	0	0	0	2	5	22	1	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
	WSAvg	7.8	8.4	7.7	7.5	6.7	6.8	6.8	6.7	7.3	8.2	8.8	9.2	9.0
(15) Precipitation	PrecAvg	864	76	56	42	52	53	78	73	87	78	102	90	77
	PrecMax	1226	163	151	120	108	98	166	154	157	189	243	233	168
	PrecMin	583	11	16	3	13	1	6	10	19	31	21	27	16
	PrecStd	150	37	35	26	25	26	32	33	34	39	52	49	43
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.6	6.2	9.0	16.2	22.8	23.7	25.3	22.7	19.6	15.1	12.5	10.2
		MCWB	5.9	5.0	6.4	10.5	16.2	16.3	18.9	18.1	16.4	13.4	12.4	9.2
	2%	DB	6.8	5.6	7.3	13.2	20.0	21.4	23.8	21.1	18.2	14.4	11.2	9.2
		MCWB	5.5	4.6	5.6	9.3	15.0	16.5	18.8	17.8	15.9	13.0	10.4	8.1
	5%	DB	6.1	5.1	6.3	11.3	18.0	19.6	22.0	20.2	17.4	13.8	10.6	8.4
		MCWB	4.9	4.4	4.9	8.3	13.8	15.9	18.6	17.4	15.3	12.6	9.8	7.3
	10%	DB	5.4	4.6	5.6	10.0	16.0	18.2	20.6	19.4	16.8	13.2	9.8	7.6
		MCWB	4.2	3.8	4.4	7.7	12.9	14.9	17.6	16.7	14.7	11.8	9.2	6.4
	0.4%	WB	6.6	5.6	6.9	11.6	16.9	18.6	20.3	19.6	17.6	14.2	12.6	9.6
		MCDB	6.9	6.0	8.1	14.9	21.7	20.7	23.5	21.1	18.6	14.4	12.6	10.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	5.7	5.1	6.1	10.1	15.5	17.3	19.5	18.7	16.9	13.6	10.9	8.7
		MCDB	6.2	5.4	7.0	12.5	18.8	19.7	22.5	20.4	17.7	14.1	11.0	9.1
	5%	WB	5.1	4.6	5.4	9.0	14.6	16.4	18.8	17.9	16.0	12.8	10.1	7.8
		MCDB	5.8	5.1	6.3	11.3	17.3	18.9	21.5	19.5	17.1	13.5	10.5	8.3
	10%	WB	4.4	4.1	4.7	7.9	13.4	15.5	18.0	17.1	15.3	12.0	9.3	6.9
		MCDB	5.2	4.6	5.5	9.8	15.7	17.8	20.2	18.8	16.5	12.9	9.9	7.6
(35) Mean Daily Temperature Range	5% DB	MDBR	3.0	2.7	2.8	3.5	3.7	3.3	3.0	2.8	2.8	2.8	2.5	3.0
		MCDBR	3.4	2.3	3.2	5.4	6.2	5.2	5.0	4.0	3.2	2.6	2.8	2.6
		MCWBR	3.6	2.4	2.4	3.3	3.7	3.5	3.1	3.1	2.7	2.6	3.0	3.2
	5% WB	MCDBR	3.2	2.4	2.9	5.1	5.6	4.3	4.4	3.2	2.6	2.3	2.7	2.5
		MCWBR	3.6	2.4	2.4	3.3	3.7	3.3	3.0	3.0	2.8	2.9	3.0	3.2
(40) Clear-Sky Solar Irradiance	taub		0.294	0.326	0.343	0.381	0.375	0.363	0.385	0.383	0.353	0.340	0.310	0.274
	taud		2.243	2.317	2.360	2.308	2.408	2.468	2.427	2.427	2.493	2.417	2.286	2.178
	Ebn at Noon		530	697	799	823	858	874	843	817	788	676	504	419
	Edh at Noon		46	69	88	109	105	101	103	96	77	63	44	35
(44) All-Sky Solar Radiation	RadAvg		0.33	0.90	2.39	3.97	5.37	6.07	5.55	4.33	2.87	1.28	0.43	0.22
	RadStd		0.06	0.17	0.23	0.52	0.47	0.35	0.55	0.26	0.28	0.18	0.09	0.02

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page